

CropSage Project Phases

Current scope, completed foundation work, accepted limitations and the path to an end-to-end crop recommendation.

PHASE 01 Foundation COMPLETE	PHASE 02 Evidence normalization NEXT	PHASE 03 Decision engine IN PROGRESS	PHASE 04 Application integration PLANNED	PHASE 05 End-to-end validation PLANNED
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Phase 1 - Farm Profile and Crop Catalog
Requirements and evidence foundations for preliminary crop suitability in Plainview, Texas.

COMPLETE

Farm Profile Foundation

- 1. Required location and planting-time inputs are defined.
- 2. Farmer irrigation availability and reliability are documented, with optional soil overrides.
- 3. Missing optional evidence reduces confidence instead of becoming a favorable match.
- 4. The demonstration farm is fixed at Plainview (34.18, -101.76) in the Texas Plains context.

Crop Catalog Foundation

- 1. Catalog version 1.1.0 contains 22 Texas crops and 29 required properties.
- 2. Temperature, soil, water, region, season and provenance fields are documented.
- 3. Nullable evidence policy prevents absent values from being treated as zero or favorable.
- 4. Every crop record remains provisional pending external agronomic review.

Location Evidence Prepared

Provider	Prepared evidence role
NASA POWER	Climate history and climatology for temperature and rainfall context.
FortyGuard	Local heat, threshold exceedance and persistence evidence.
Open-Meteo	Current, recent and forecast weather plus modeled soil moisture.
USDA SSURGO	Plainview soil properties, root-zone limits and available water storage; unified evidence exported.

Phase 1 outcome

The Plainview MVP now has a defined farm profile, a complete crop-requirement catalog and mapped evidence responsibilities. Production normalization and scoring integration remain next.

Accepted MVP Limitations

These constraints are intentional project decisions. They should remain visible in implementation, testing and user-facing interpretation.

1	Plainview is manually assigned to the Texas Plains; no statewide region resolver.
2	Frost-free-season comparison is unavailable and must remain unknown or excluded.
3	Heat thresholds, duration and numeric water ranges remain supplementary under the new policy.
4	Records remain provisional pending external agronomic review.
5	Exact farm boundaries, geocoding, well/canal capacity and farmer soil-depth overrides are optional.
6	Recommendations apply to preliminary suitability, not yield or irrigation prescriptions.

Remaining Implementation Work

PHASE 2	Fix and generate normalized NASA and Open-Meteo evidence exports. Preserve or regenerate sanitized FortyGuard evidence. Combine provider outputs into an evidence bundle.
Evidence normalization	
PHASE 3	Finish the deterministic engine.
Decision engine	
PHASE 4	Build production adapters, application flow and UI.
Product integration	
PHASE 5	Run the final end-to-end Plainview recommendation.
End-to-end validation	

NEXT MILESTONE	A validated, normalized evidence bundle ready for deterministic scoring.
MVP FINISH LINE	A reproducible Plainview run that produces preliminary crop suitability recommendations with reasons, risks and confidence.

Document basis: approved CropSage MVP decisions and current Plainview implementation status.

Database Implementation Decisions

A statewide-ready persistence layer for farm profiles, provider evidence, deterministic recommendations and validated application state. Engine contracts are received and ranking semantics locked.

PHASE STATUS	IMPLEMENTED LOCALLY	Migrations 1-6 pass 212 database checks; the corrected ranking fixture remains the final integration acceptance task.
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Decision Register 1-8

Status	Decision	Implementation note
SKIPPED	1. Texas location strategy	The database will not resolve Texas regions. API and application adapters supply resolved region and timezone values; the database stores their method and version.
LOCKED	2. Engine input and output contracts	FarmProfile, EvidenceBundle, scoring configuration, 22-crop recommendation and validation samples received. Their schemas now drive migrations 4-5.
DURING BUILD	3. Exact database column definitions	Types, nullability, defaults, enums, checks and foreign-key actions will be locked in each migration and reflected in tests.
LOCKED	4. Consistent naming	Use recommendation_run_id for run foreign keys and external_profile_id for the application-level profile identifier. Apply one name consistently in SQL and adapters.
LOCKED	5. Evidence origins	Evidence records identify source_type, source_name and source_metadata_jsonb so provider, farmer, laboratory and derived values remain distinguishable.
LOCKED	6. Derivation lineage	Derived evidence stores derivation_jsonb with calculation name, version and input evidence-record IDs.
LOCKED	7. Provider evidence separation	Provider artifacts and normalized records stay catalog-independent. EvidenceBundle may include catalog-versioned crop evidence; factors, caps and scores remain engine output.
LOCKED	8. Artifact storage	Use artifacts_jsonb arrays on provider_fetches and recommendation_runs for Storage paths, checksums, content types, sizes and schema versions. No additional artifact table for the MVP.

Persistence, Security And Integrity

These decisions govern state transitions, deterministic replay, ranking, provider retries, access control and spatial data.

Decision Register 9-16

Phase	Decision	Implementation Note
LOCKED	9. Lifecycle statuses	Use explicit state transitions. Runs follow pending -> running -> scored -> validated -> completed, with failed available from active states. Rows become immutable at their terminal state.
LOCKED	10. Deterministic hashes	Store SHA-256 hashes for canonical engine inputs, outputs, evidence bundles and artifacts. Canonical JSON rules are implemented and contract-tested.
LOCKED	11. Ranking semantics	Store regionally_eligible, overall_rank (1-22) and nullable eligible_rank. Eligible crops appear first; ineligible crops retain overall rank but eligible_rank = null.
LOCKED	12. Provider retries and caching	Each provider_fetches row represents one HTTP attempt. request_group_id groups attempts; cache_key finds a reusable successful result; failed attempts remain auditable.
LOCKED	13. Authentication and RLS	The browser calls the application server. Supabase privileged credentials remain server-side; browser roles receive no direct writes; RLS and grants protect every exposed table.
LOCKED	14. Foreign keys and indexes	Create ownership foreign keys, explicit deletion behavior and indexes for foreign keys, RLS joins, cache lookup, ranking, time retrieval and spatial access.
LOCKED	15. Spatial representation	Store farm_point as PostGIS Point and optional farm_boundary as normalized MultiPolygon, both SRID 4326. Resolution metadata remains separate.
SKIPPED	16. Database handling of provisional catalog status	Do not add a separate database review-status gate in this phase. The Git catalog remains authoritative; runs still preserve catalog version and hash.
CONTRACTS RECEIVED	Database implementation is cleared to start. The engine must regenerate its ranking fixture before scoring integration acceptance, using overall_rank and nullable eligible_rank.	

Supabase Build Sequence

Migrations are applied locally first, verified through reset and pgTAP, then pushed to the linked Supabase project after review.

#	Migration group	Deliverables
1	Bootstrap	Initialize Supabase; configure migrations, tests and seed files; enable PostGIS and required extensions; define enums and helpers.
2	Sessions and profiles	Create assessment_sessions and farm_profiles with canonical naming, Point/MultiPolygon geography, versioning, ownership and initial RLS.
3	Provider collection	Create provider_fetches, retry/cache constraints, typed request metadata, artifacts_jsonb and the private provider-artifacts bucket.
4	Evidence and bundles	Create evidence_records, origins and lineage; finalize evidence_bundles and membership from the received 1.1.0 contract; preserve exact bundle JSONB and hash.
5	Scoring persistence	Create recommendation_runs and crop_score_results using overall_rank and eligible_rank; store validation reports, render authorization and deterministic hashes.
6	Operations and hardening	Create run_events; add RLS grants, append-only and terminal immutability triggers, foreign-key indexes, server-only Storage controls and complete database tests.

Acceptance Gates

1 Local supabase db reset recreates the complete schema from migrations and seed files.	5 Provider, farmer, laboratory and derived evidence retain source and lineage metadata.
2 Anonymous and authenticated browser roles cannot directly write protected application tables or Storage objects.	6 An EvidenceBundle round-trips with the same canonical SHA-256 hash.
3 A farm profile preserves version, resolved location metadata and canonical spatial values.	7 A corrected engine fixture stores 22 crop rows, eligible crops first, unique overall ranks and null eligible_rank for ineligible crops.
4 Provider retries remain distinct while successful cache reuse is deterministic.	8 Validation alone authorizes completed presentation state; terminal records reject ordinary update and delete operations.

PHASE OUTCOME	A reproducible Supabase schema with secure farm state, auditable evidence, deterministic run persistence and migration tests, ready for application adapters and statewide Texas recommendation runs.
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Frontend Architecture Decisions

A responsive farmer decision interface backed by Next.js, a server-only application boundary, the Python scoring engine and the implemented Supabase persistence model.

PHASE STATUS	READY TO START	All frontend defaults are approved. The corrected engine fixture is required only for final scoring integration acceptance.
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Decision Register

#	Decision	Implementation rule
1	Application framework	Use Next.js with TypeScript and App Router under apps/web.
2	Browser/server boundary	The browser calls application endpoints only. Provider, model and privileged Supabase credentials remain server-side.
3	Scoring service	Expose the deterministic Python engine through a typed FastAPI contract. Next.js acts as the browser-facing backend-for-frontend.
4	Session model	Use silent anonymous Supabase sessions with no login page. Accounts and cross-device history remain outside the MVP.
5	Location input	Accept a Texas map pin or manual coordinates and provide a Plainview demo preset. No address search or field-boundary drawing.
6	Scenario interaction	Use structured planting, irrigation and supported farm-evidence controls. Do not store unrestricted chat history.
7	Progress delivery	Use server-sent events with polling fallback. Display only user-safe status text; run_events remains the persisted audit trace.
8	Result authority	Render scores only after validation_reports.render_allowed is true. The UI never recalculates or modifies engine values.

PRODUCT LANGUAGE	Use CropSage, preliminary suitability and consider. Never present yield guarantees, irrigation prescriptions or instructions to plant.
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Interface Map

Five focused interfaces cover the complete farmer workflow without adding a marketing page, database dashboard or catalog-management surface.

Route	Interface	Primary behavior	Data contract
/farm	Farm Setup	Location, planting plan, irrigation, optional field evidence and final review.	assessment_sessions farm_profiles
/assessments/[id]/progress	Analysis Progress	Bounded provider, evidence, scoring and validation activity with live/cache/failure states.	provider_fetches evidence_bundles run_events
/assessments/[id]/results	Recommendation Results	Top-three comparison plus all 22 crops, with eligible and regionally unsuitable groups separated.	recommendation_runs crop_score_results validation_reports
/assessments/[id]/crops/[cropId]	Crop Details	Summary, factor breakdown, requirements-versus-evidence and source/freshness details.	crop_score_results evidence_records Git crop catalog
/assessments/[id]/scenarios	Scenario Comparison	Change supported inputs, confirm assumptions and compare baseline versus scenario deltas.	recommendation_runs parent_run_id scenario deltas

Presentation Contracts

RESULTS	Show suitability, confidence and risk separately. Missing evidence is unknown, never zero or favorable.
RANKING	Display all 22 crops. Eligible crops use eligible_rank; unsupported crops retain overall_rank in a separate group.
EVIDENCE	Show provider, timestamp, freshness, spatial resolution and live/cache/fallback state without exposing raw payloads.
VALIDATION	A rejected or incomplete validation report produces a blocked-results state, not partially rendered scores.
RESPONSIVE	Use a compact comparison table on desktop and stacked, non-overlapping sections on mobile; crop details become a full mobile page.

FORM BOUNDARY	Farmers enter location, planting and optional field evidence. Region, timezone, crop root-zone depth and SSURGO water capacity are resolved by application, catalog and provider layers.
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Frontend Build Sequence

Build against typed mock contracts first, then replace mocks with server adapters and the corrected deterministic-engine fixture without changing presentation rules.

#	Implementation group	Deliverable
1	Application foundation	Create apps/web; configure Next.js, TypeScript, responsive tokens, accessible controls, Supabase session bootstrap and typed API DTOs.
2	Farm Setup	Build the five-step form, Texas map/manual coordinate input, Plainview preset, validation, unknown-value handling and profile review.
3	Analysis Progress	Build bounded progress states for four providers, evidence assembly, scoring and validation using mock server events.
4	Recommendation Results	Build validated top-three comparison, complete 22-crop ranking, eligibility grouping, confidence, risks and blocked-render state.
5	Crop Details and evidence	Add factor breakdown, requirement/evidence comparisons, provenance, freshness and user-safe warnings.
6	Scenario Comparison	Add structured changes, assumptions confirmation, parent-run context and score/rank deltas.
7	Production integration	Connect Next.js server routes, FastAPI scorer, Supabase persistence and event streaming; verify failure, cached fallback, mobile and accessibility states.

Acceptance Gates

1	Required farm inputs are enforced; optional evidence can remain explicitly unknown.	5	Only validator-authorized results render; rejected results remain blocked.
2	Any Texas point can be submitted and Plainview can be loaded as the demo preset.	6	All 22 crops preserve eligibility-first ranking and separate ineligible presentation.
3	The browser receives no provider, model or privileged Supabase credentials.	7	Scenario results identify their baseline and show unchanged assumptions plus deltas.
4	Progress always reaches success, fallback or actionable failure; no indefinite loading.	8	Desktop and mobile views have keyboard access, visible labels and no overlap or clipped text.

INTEGRATION DEPENDENCY	The corrected engine output, recommendation schema and validation report are required before Group 4 production wiring and final acceptance, but Groups 1-3 can begin immediately.
PHASE OUTCOME	A usable Texas farm-planning flow that collects a trusted profile, shows bounded evidence activity, renders validated 22-crop recommendations and compares reproducible scenarios.